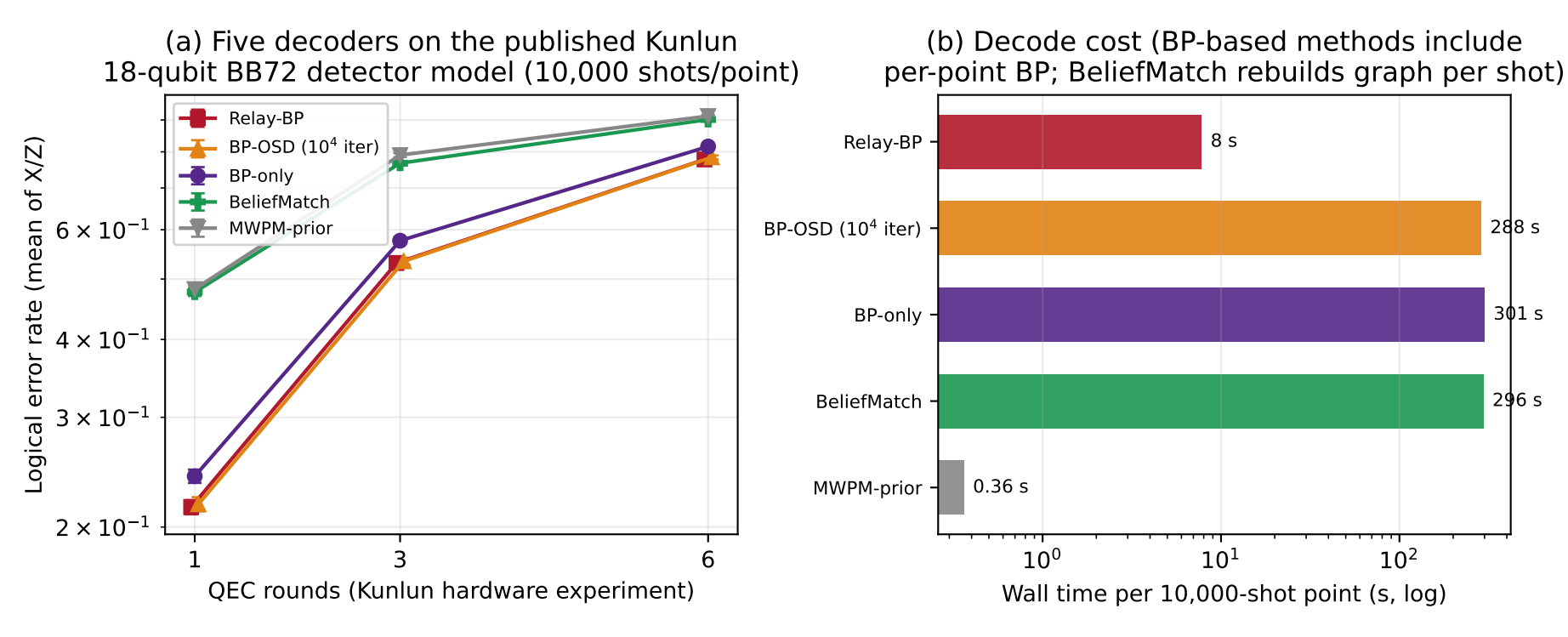
(a) Five decoders on the published Kunlun
18-qubit BB72 detector model (10,000 shots/point)
Relay-BP
BP-OSD (10^4 iter)
BP-only
BeliefMatch
MWPM-prior
Logical error rate (mean of X/Z)
6×10^{-1}
4×10^{-1}
3×10^{-1}
2×10^{-1}
1
3
6
QEC rounds (Kunlun hardware experiment)
(b) Decode cost (BP-based methods include
per-point BP; BeliefMatch rebuilds graph per shot)
Relay-BP
BP-OSD (10^4 iter)
BP-only
BeliefMatch
MWPM-prior
8 s
288 s
301 s
296 s
0.36 s
10^0
10^1
10^2
Wall time per 10,000-shot point (s, log)